ENGINEERING SPECIFICATION • SYSTEM INTEGRATIONS

MyCA: Production Integration Blueprint

Architectural blueprint for connecting MyCA's deterministic books, GST reconciliation, and defensible audit engines to live statutory portals, licensed GSPs, banking networks, and firm infrastructure.

SYSTEM CORE	SOURCE OF TRUTH	DATA PROTECTION	AUTONOMOUS LIMIT
TigerBeetle Invariants	Tax Invoice Entry	DPDPA 2023 India Only	Zero Live Portal Filing

The Adapter Invariant

External connectors are strict protocol translators. Remote payloads (from Quicko, GSTN, Tally, or Banks) are ingested and converted into existing MyCA schemas. Under no circumstances does an external API connector replace or duplicate internal accounting or reconciliation logic. All statutory uploads, OTPs, and signatures require authenticated CA approvals.

01. Statutory Tax & Government Portals

Direct and GSP-mediated interfaces for GSTN, NIC e-Invoice, Income Tax e-Filing, and NSDL/TRACES.

REQUIREMENT	SYSTEM & SCOPE	TARGET PROVIDER & DOCS	DIRECTION	APPROVAL GATE
REQ-CA-006	**GSTN via Licensed GSP** Auth, return upload (1/3B), ledger balance, IMS, GSTR-2B job polling. `POST /gst/auth • POST /gst/gstr1`	Quicko Sandbox GST API ↗ Official GSTN Portal ↗	Bi-directional	CA OTP Required
REQ-CA-007	**NIC E-Invoice & E-Way Bill** Official v1.04 auth with RSA PKCS#1 v1.5 public key encryption; IRN generation. `POST /eivital/v1.04/auth • POST /Invoice`	NIC Developer Sandbox ↗ Sandbox E-Invoice Guide ↗	Bi-directional	CA Authorized
REQ-CA-008	**GSTIN Status & Return Tracking** Legal name verification, state registration, composition check, return history. `GET /gstin/{gstin}`	Sandbox Return Tracking API ↗	Read-Only	Autonomous
REQ-CA-009	**AIS & Form 26AS Import** Annual Information Statement financial transactions, high-value purchases, TDS credits. `POST /income-tax/ais/import`	ITD e-Filing Portal ↗ Quicko Tax Analytics ↗	Read-Only	Autonomous
REQ-CA-010	**TDS Returns & TRACES** Deductee payments, Section 194A/C/I/J thresholds, challan mapping, FVU generation. `GET /tds/payments • POST /fvu`	TRACES CPC Portal ↗ Protean TIN NSDL ↗	Bi-directional	CA Review Gate

02. Accounting Software & Financial Feeds

Bi-directional synchronization with client books and automated daily bank statement streaming.

PLATFORM	PROTOCOL & LOCAL INTERFACE	OFFICIAL DOCUMENTATION	CAPABILITIES
TallyPrime Live Sync Direct local desktop sync.	XML over HTTP Request Envelope ODBC Port 9000 `http://127.0.0.1:9000`	Tally XML Data Interchange ↗	Real-time Day Book sync preserving cost centres and voucher types.
Zoho Books Connector Cloud accounting integration.	REST API v3 (OAuth 2.0) JSON Payloads `api.zoho.com/books/v3`	Zoho Books API Reference ↗	Auto-streams sales invoices, bills, and payment settlements.
Account Aggregator (AA) RBI-regulated bank feeds.	Financial Information User (FIU) Consent Artifact Validation `Encrypted FI Data Blocks`	Sahamati AA Architecture ↗ Setu Account Aggregator ↗	Daily bank statement feeds with automated running-balance tie-out.

03. Practice Automation & Client Communication

Automated document collection and multi-channel compliance reminder dispatch.

REQUIREMENT	WORKFLOW	PROVIDER & TRANSPORT	FLOW	CONTROL POLICY
`REQ-CA-011`	**Tasks & Statutory Deadlines** Dynamic calendar accounting for gazetted holidays and weekend rollovers.	Practice Management Database (Finexo / Karbon / Custom) `GET /tasks • POST /tasks`	Read / Write	Automated
`REQ-CA-012`	**Client Document Chasing** Tracking missing bank statements, bills, and GST certificates.	Secure Client Upload Portal Webhook Status Tracking `POST /document-requests`	Bi-directional	Automated
`REQ-CA-013`	**Client Notifications** Filing alerts and document collection reminders via WhatsApp/Email.	Meta WhatsApp Cloud API ↗ AWS SES Transactional Email ↗	Outbound	Template Approval

04. Firm Master Data & Core Infrastructure

Practice-wide client master, multi-tenant ledger storage, and immutable audit logs.

REQUIREMENT	COMPONENT	PRODUCTION STORAGE ARCHITECTURE	INTEGRITY & COMPLIANCE
REQ-CA-001	**Client Master Registry** Entity type, PAN, GSTINs, turnover, GST filing frequency, audit mandate.	PostgreSQL / Firm CRM REST Endpoint: `GET /clients`	Single truth source across all firm compliance engagements.
REQ-CA-002	**Multi-Client Ledger Database** Distributed storage replacing single local state.json.	TigerBeetle Financial Database ↗ Distributed state machine / PostgreSQL fallback	Integer paise BigInt, balanced transfers, CRC-32 chain. Tested at 5,700 tx / 72ms.
REQ-CA-003	**Encrypted Document Archive** Tally XMLs, bank PDFs, GST JSONs, working papers.	AWS S3 / Azure Blob (India Region) Server-Side AES-256 Encryption	DPDPA 2023 compliance with strict India data residency guarantees.
REQ-CA-004	**Firm Users & RBAC** Maker-checker approval hierarchies.	Microsoft Entra ID (Azure AD) ↗ OAuth 2.0 / SAML 2.0 SSO	Separation of duties: Article Clerk • Reviewer • Partner Sign-off.
REQ-CA-005	**Tamper-Evident Audit Log** Chronological ledger of tool runs and state transitions.	WORM Storage / Append-Only Ingestion SHA-256 Digest Chains	Defensible audit papers satisfying ICAI Standard on Auditing (SA) 230.

05. Hardware Security, Cryptography & AI Telemetry

Hardware-bound digital signatures, OS credential vaulting, and AI grounding pipelines.

SUBSYSTEM	TECHNOLOGY & STANDARDS	OFFICIAL REFERENCE	SECURITY MODEL
Digital Signature (DSC) Cryptographic return signing.	PKCS#11 Native USB Token Driver ePass2003 / Watchdata / ProxKey	Controller of Certifying Authorities ↗	Physical Token + PIN Zero autonomous signing
OS Credential Vault Securing local API secrets.	Windows DPAPI (`CryptProtectData`) Hardware TPM-backed entropy	Microsoft DPAPI Architecture ↗	OS-Level Encrypted Replaces plaintext govt.json
REQ-CA-014: Knowledge Base AI statutory grounding.	Vector Database RAG Pipeline Qdrant / pgvector Indexed Laws	Qdrant Vector Database ↗	Grounds queries in verified circulars, IT Act, and firm SOPs.
REQ-CA-015: Quality Evals LLM correction logging.	Telemetry & Correction Tracking Langfuse / Dedicated Eval Store	Langfuse Engineering Platform ↗	Tracks reviewer corrections to continuously fine-tune prompt routing.

06. Recommended Implementation Sequencing

Milestone 1: Practice Data Layer
- Central PostgreSQL / TigerBeetle database (REQ-CA-001 & 002)
- Encrypted S3 Document Vault (REQ-CA-003)
- Windows DPAPI credential migration

Milestone 2: Statutory Connectors
- Quicko Sandbox GSP ASP onboarding (REQ-CA-006)
- Live GSTR-2B ingestion and IRN e-invoicing (REQ-CA-007)
- GSTIN live validation API (REQ-CA-008)

Milestone 3: Client Automation
- Meta WhatsApp Business Cloud API integration (REQ-CA-013)
- Automated monthly document chase portal (REQ-CA-012)
- Holiday-aware deadline dispatcher (REQ-CA-011)

Milestone 4: Direct Accounting Sync
- TallyPrime Port 9000 live XML sync
- Zoho Books OAuth 2.0 bi-directional adapter
- Account Aggregator automated daily bank statements